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SYSTEM AND METHOD FOR GENERATING MEDIA PRESENTATION WITH ZOOM-IN EFFECT

Abstract

A method and system for generating a media presentation. The method includes receiving a first image, from a media objects comprising a specified object. The method includes identifying a region of interest (ROIs) in the first image based on the specified object. Further, the method includes determining a zoomable ROI from the identified ROIs, with a zoom-in effect. The method includes retrieving a consecutive image from the media objects based on the zoomable ROI. Further, the method includes generating the consecutive image based on the zoomable ROI based on a failure to retrieve the consecutive image from the media objects and embedding the consecutive image with the first image to generate the media presentation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2023/016314 designating the United States, filed on Oct. 20, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Indian Patent Application No. 2022211072049, filed on Dec. 13, 2022, in the Indian Patent Office, the disclosures of each of which are incorporated by reference herein in their entireties.

BACKGROUND

[0002] The disclosure relates to stitching together media presentation and for example relates to system and method generating media presentation in relevant sequence with zoom-in effect.

DESCRIPTION OF RELATED ART

[0003] With the progress in content development, creators are in requirement of developing engaging content while reducing efforts to do the same. It may be required that engaging content offers something new to a viewer such as a new perspective, an unexpected laugh, some knowledge, or something helpful, inspiring, or entertaining. Engaging content may provide the viewer a peek at something they haven't seen before but can relate to in some way.

[0004] The existing techniques require the creator to put in manual efforts for editing the content to create new experiences, such as transitioning content. Such manual efforts may be time and resource consuming. In an example to create engaging content with media files such as images, the user may create visual transitions. In visual transition, animation may introduce a new visual component to the scene or existing content by replacing a component or media file already in the scene. Further, the user may require a special effect such as zooming into a current media file which may be currently visible to the viewer or any component in the current media file before transitioning to a consecutive media file.

[0005] In existing technologies, the creator may have to manually scheme through multiple media files to arrange them in sequential order to create a zoomable effect transition between multiple media files. The zoomable effect transition between media files may create a zooming in-effect for the viewer, wherein the component in the current media file is zoomed-in and subsequently the consecutive media file appears. It may be required that the consecutive media file is contextually relevant to the current media file such that the viewer may experience zooming-in on the component and the consecutive media file out of the zoomed-in component.

[0006] Creating such zooming-in transitioning effect in the media presentation may require considerable time and manual efforts by the creator. The creator may have to manually arrange the media files in sequential order so that contextual relevance is maintained. Moreover, the creator may need to identify a region of interest in each of the media file which may be zoomed-in to present the consecutive media file. In an event the creator does not have contextually relevant image, the zooming-in transition effect may not provide relevant experience and may seem inappropriate.

SUMMARY

[0007] According to an example embodiment of the present disclosure, a method for generating a media presentation is disclosed. The method includes: receiving a first image, from a plurality of media objects comprising a plurality of specified objects; identifying a plurality of regions of interest (ROIs) in the first image based on the specified objects; determining at least one zoomable ROI from the identified plurality of ROIs, wherein the at least one zoomable ROI includes one of the plurality of ROIs with a zoom-in effect; retrieving a consecutive image from the plurality of media objects based on the at least one zoomable ROI; and generating the consecutive image based on the at least one zoomable ROI based on failure to retrieve the consecutive image from the plurality of media objects and embedding the consecutive image with the first image to generate the media presentation.

[0008] According to an example embodiment of the present disclosure, a system for generating a media presentation is disclosed. The system includes: an identification module comprising circuitry configured to receive a first image, from a plurality of media objects comprising a plurality of specified objects and identify a plurality of regions of interest (ROIs) in the first image based on the specified objects; a determining module comprising circuitry configured to determine at least one zoomable ROI from the identified plurality of ROIs, wherein the at least one zoomable ROI includes one of the plurality of ROIs with a zoom-in effect; a retrieving module comprising circuitry configured to retrieve a consecutive image from the plurality of media objects based on the at least one zoomable ROI; and a generating module comprising circuitry configured to generate the consecutive image based on the at least one zoomable ROI based on failure to retrieve the consecutive image from the plurality of media objects and embed the consecutive image with the first image for generating the media presentation.

[0009] To further clarify the advantages and features of the disclosure, a more detailed description will be provided with reference to various example embodiments thereof, which are illustrated in the appended drawings. It is appreciated that these drawings depict example embodiments and are therefore not to be considered limiting of its scope. The disclosure will be described and explained with additional specificity and detail with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other features, aspects, and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings in which like characters represent like parts throughout the drawings, and in which:

[0011] FIG. 1 is a diagram illustrating an environment for the implementation of a system for generating a media presentation, according to various embodiments;

[0012] FIG. 2A is a block diagram illustrating an example configuration of a system for generating the media presentation, according to various embodiments;

[0013] FIG. 2B is a diagram illustrating example modules/software components of a system for generating the media presentation, according to various embodiments;

[0014] FIG. 3 is a diagram illustrating an example process flow of identifying a region of interest in an image by an identification module of the system, according to various embodiments;

[0015] FIG. 4 is a diagram illustrating an example process flow of determining the region of interest in the image by a determining module of the system, according to various embodiments;

[0016] FIG. 5 is a diagram illustrating an example process flow for determining a geometric feature score for the region of interest in the image by a geometric feature sub-module of the system, according to various embodiments;

[0017] FIG. 6 is a diagram illustrating an example process flow for determining an image depth score for the region of interest by an image depth score sub-module of the system, according to various embodiments;

[0018] FIG. 7 is a diagram illustrating an example process flow for determining a zoomable score for the region of interest by a zoomable score sub-module of the system, according to various embodiments;

[0019] FIG. 8 is a diagram illustrating an example process flow for retrieving a consecutive image by a retrieving module of the system, according to various embodiments;

[0020] FIG. 9A is a diagram illustrating an example process flow of a dynamic global-local attention sub-module for determining a dynamic global local vector in the generating module of the system, according to various embodiments;

[0021] FIG. 9B is a diagram illustrating an example process flow for generating the consecutive image by the generating module of the system, according to various embodiments;

[0022] FIG. 10 is a diagram illustrating an example use-case for generating the media presentation, according to various embodiments; and

[0023] FIG. 11 is a flowchart illustrating an example method for generating the media presentation, according to various embodiments.

[0024] Those skilled in the art will appreciate that elements in the drawings are illustrated for simplicity and may not have necessarily been drawn to scale. For example, the flowcharts illustrate the method in terms of steps involved to help to improve understanding of aspects of the disclosure. Furthermore, in terms of the construction of the device, one or more components of the device may have been represented in the drawings by conventional symbols, and the drawings may show those specific details that are pertinent to understanding various embodiments of the disclosure so as not to obscure the drawings with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0025] Reference will now be made to various example embodiments and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended, such alterations and further modifications in the illustrated system, and such further applications of the principles of the disclosure as illustrated therein being contemplated as would occur to one skilled in the art to which the disclosure relates.

[0026] It will be understood by those skilled in the art that the foregoing general description and the following detailed description are explanatory of the disclosure and are not intended to be restrictive thereof.

[0027] Reference throughout this disclosure to “an aspect,” “another aspect” or similar language may refer, for example, to a particular feature, structure, or characteristic described in connection with an embodiment being included in at least one embodiment of the disclosure. Thus, appearances of the phrase “in an embodiment,” “in another embodiment” and similar language throughout this disclosure may, but do not necessarily, all refer to the same embodiment.

[0028] The terms “comprises,” “comprising,” or any other variations thereof, are intended to cover a non-exclusive inclusion, such that a process or method that comprises a list of steps does not include only those steps but may include other steps not expressly listed or inherent to such process or method. Similarly, one or more devices or sub-systems or elements or structures or components preceded by “comprises . . . a” does not, without more constraints, preclude the existence of other devices or other sub-systems or other elements or other structures or other components or additional devices or additional sub-systems or additional elements or additional structures or additional components.

[0029] FIG. 1 is a diagram illustrating an example environment for the implementation of a system **100** for generating a media presentation **104**, according to various embodiments. For the sake of brevity, the system **100** for generating the media presentation **104** is hereinafter interchangeably

referred to as the system **100**.

[0030] In an embodiment, referring to FIG. **1**, the system **100** may be implemented in a user equipment (UE) **102**. In an example, the UE **102** may be but not limited to, a laptop computer, a desktop computer, a Personal Computer (PC), a notebook, a smartphone, a tablet, a smart watch and the like.

[0031] In an embodiment, the system **100** may be implemented between a user (now shown) and the UE **102**. In an example, the user desire to generate a media presentation **104** in the UE **102**. The media presentation **104** is indicative of creating a slideshow between a plurality of media files such as images. The slideshow may include transitioning between the images with a zooming effect on a first image **106**. The first image **106** may be the earliest image from where the slideshow initiates. In continuing the slideshow, a consecutive image is displayed, and the process may be repetitive to display the consecutive image one after another with an image continuity sequence and a semantic relevance between images.

[0032] While displaying the consecutive image in the form of the media presentation **104**, a zoom-in effect is created in each of the image upon presentation. The zoom-in effect is created on an identified region of interest (ROI) **112a**. There may be multiple ROIs **112** in the image. In an example, the first image **106** may have multiple ROIs **112** such as ROI-1, ROI-2, ROI-3, ROI-n. The system **100** may identify a zoomable ROI **112a** from the multiple ROIs **112**. The zoomable ROI **112a** may then be selected for applying zoom-in effect before displaying the consecutive image **114**. Such that application of zoom-in effect may create a transition illustrating that the consecutive image **114** originates from the zoomable ROI **112a**.

[0033] As illustrated in FIG. **1**, in an example, the first image **106** may be selected by the user from a database **102a** of the UE **102** or alternatively the user may capture the first image **106** using a camera of the UE **102** such that the UE **102** may be a source of the first image **106**. The database **102a** may include storing a media object **108** such as a cluster of images stored in the database **102a**. In the example, the media object **108** may provide the first image **106** and any consecutive image(s) **114**. The database **102a** may also include a predefined (e.g., specified) object **110** corresponding to the media object **108**. In the example, the predefined object **110** may be known set of objects present in the media object **108**. In the example, the first image **106** may include plurality of the pre-defined objects **110** such as a sun, a house, a tree. A dotted line bounding box is created on each of the pre-defined (e.g., specified) objects **110** in the first images identifying them as multiple ROIs.

[0034] In an embodiment, the zoomable ROI **112a** is determined from the multiple ROIs in the first image **106**. In an example, a window in the house, in the first image **106** is determined. A zoom-in effect is applied on the window being the zoomable ROI **112a** and subsequently the consecutive image **114** is displayed. In the example, the consecutive image **114a** may be displayed after zooming in on the zoomable ROI **112a** e.g., the window. The consecutive image **114a** may represent a room set-up which may contextually be relevant to the first image **106** and the zoomable ROI **112a**. In this instance, the window is selected as the zoomable ROI **112a** and the zoom-in effect may be applied on the window, thus it may be contextually relevant to display the room set-up depicted in the consecutive image **114a**. Similarly, now the zoomable ROI **112a** is determined in the consecutive image **114a** being the room scenario. In the example, the zoomable ROI **112a** in the consecutive image **114a** may be determined as a river scene in a painting present in the room scenario. Thus, the zoom-in effect is applied on the zoomable ROI **112a** e.g., the river scene and the consecutive image **114b** is displayed. In the example, the consecutive image **114b** displaying a fish in river may be contextually relevant as the painting in the previous consecutive image **114a** displays the river scene. Such process of determining the zoomable ROI **112a** in each image and presenting the consecutive image **114** continues until the user ends transition, the media objects **108** in the database **102a** is exhausted, or contextually the system **100** may cease to generate any consecutive image **114**.

[0035] The user may provide instructions to the UE **102** for generating the media presentation **104** via an application installed in the UE **102**. The application may be used for receiving instructions from the user and generating the media presentation **104** using a processor of the system **100**.

[0036] FIG. 2A is a block diagram illustrating an example configuration of the system **100** for generating the media presentation **104** according to various embodiments. According to an embodiment, the system **100** includes at least one or more processors (e.g., including processing circuitry) **202**, a memory **204**, a module/unit (e.g., including various circuitry and/or executable program instructions executed on the circuitry) **206**, the database **102a**, an audio/video (AV) unit (e.g., including A/V circuitry) **209**, a Network Interface (NI) **211** coupled with each other. The system **100** may be further coupled with a server **215** via an internet **213** (not shown).

[0037] FIG. 2B is a diagram illustrating example modules/software components of the system **100** for generating the media presentation **104** according to various embodiments.

[0038] In an embodiment, referring to FIGS. 2A and 2B, the UE **102** may include, but is not limited to, a processor (e.g., including processing circuitry) **202**, memory **204**, modules (e.g., including various circuitry and/or executable program instructions executed on various circuitry) **206**, and database **102a**. The modules **206** and the memory **204** may be coupled to the processor **202**.

[0039] The processor **202** may include various processing circuitry, for example, a single processing unit or several units, all of which could include multiple computing units. The processor **202** may be implemented as one or more microprocessors, microcomputers, microcontrollers, digital signal processors, central processing units, state machines, logic circuitries, and/or any devices that manipulate signals based on operational instructions. Among other capabilities, the processor **202** is adapted to fetch and execute computer-readable instructions and data stored in the memory **204**. At this time, one or a plurality of processors may be a general purpose processor, such as a central processing unit (CPU), an application processor (AP), or the like, a graphics-only processing unit such as a graphics processing unit (GPU), a visual processing unit (VPU), and/or an AI-dedicated processor such as a neural processing unit (NPU). The one or a plurality of processors control the processing of the input data in accordance with a predefined operating rule or artificial intelligence (AI) model stored in the non-volatile memory and the volatile memory. The predefined operating rule or artificial intelligence model is provided through training or learning. Further, the processor **202** may include various processing circuitry and/or multiple processors. For example, as used herein, including the claims, the term “processor” may include various processing circuitry, including at least one processor, wherein one or more of at least one processor, individually and/or collectively in a distributed manner, may be configured to perform various functions described herein. As used herein, when “a processor”, “at least one processor”, and “one or more processors” are described as being configured to perform numerous functions, these terms cover situations, for example and without limitation, in which one processor performs some of recited functions and another processor(s) performs other of recited functions, and also situations in which a single processor may perform all recited functions. Additionally, the at least one processor may include a combination of processors performing various of the recited/disclosed functions, e.g., in a distributed manner. At least one processor may execute program instructions to achieve or perform various functions.

[0040] A detailed explanation of each of the aforesaid component as shown in FIGS. 2A and 2B will be explained in greater detail below. Further, the working of the system **100** will be explained with respect to FIGS. 2A and 2B. The reference numerals are kept the same in the disclosure wherever applicable for the ease of explanation.

[0041] The memory **204** may include any non-transitory computer-readable medium known in the art including, for example, volatile memory, such as static random-access memory (SRAM) and dynamic random-access memory (DRAM), and/or non-volatile memory, such as read-only memory (ROM), erasable programmable ROM, flash memories, hard disks, optical disks, and magnetic

tapes.

[0042] The modules **206**, amongst other things, may include various circuitry and/or routines, programs, objects, components, data structures, etc., which perform particular tasks or implement data types and may be executed by the various circuitry. The modules **206** may also be implemented as, signal processor(s), state machine(s), logic circuitries, and/or any other device or component that manipulates signals based on operational instructions.

[0043] The modules **206** can be implemented in hardware, instructions executed by a processing unit, or by a combination thereof. The processing unit can comprise a computer, a processor (see, e.g., description above), a state machine, a logic array, or any other suitable devices capable of processing instructions. The processing unit can be a general-purpose processor which executes instructions to cause the general-purpose processor to perform the required tasks or, the processing unit can be dedicated to performing the required functions. In an embodiment of the present disclosure, the processor **202** includes the modules **206** and the processor **202** via the modules **206** is configured to execute machine-readable instructions (software) which perform the working of the system **100** within the scope of the disclosure as described in forthcoming paragraphs.

[0044] In an embodiment, the modules **206** may include an image capturing module **203**, an identification module **210**, a determining module **212**, a retrieving module **214**, and a generating module **216**. The identification module **210**, the determining module **212**, the retrieving module **214**, and the generating module **216** may be in communication with each other. The data **208** serves, amongst other things, as a repository for storing data processed, received, and generated by one or more of the modules **206**.

[0045] Referring to FIGS. **1**, **2A** and **2B**, the image capturing module **203** is configured to capture the first image **106** via a camera installed in the UE **102**. Alternatively, the image capturing module **203** is in communication with the database **102a** and is configured to retrieve the first image **106** which may be stored in the media objects **108**.

[0046] In an embodiment, the identification module **210** is configured to receive the first image **106** from the media objects **108** or from the image capturing module **203**. The first image **106** may include a pre-defined objects **110**. Further, the identification module **210** is configured to identify multiple ROIs **112** e.g., ROI-1, ROI-2, ROI-3, ROI-n in the first image **106** based on the pre-defined objects **110**. The identification module **210** is in communication with the determining module **212**.

[0047] In an embodiment, the determining module **212** may be configured to determine the zoomable ROI **112a** among the identified multiple ROIs **112**. The determining module **212** is configured to determine pre-defined object's shape and state score such as zoomable score to determine the zoomable ROI **112a** from the identified multiple ROIs **112** in the first image **106**. In an example, the zoom-in effect may be applied on the zoomable ROI **112a**. The identification module **210** and the determining module **212** is in communication with the retrieving module **214**.

[0048] In an embodiment, the retrieving module **214** may be configured to retrieve the consecutive image **114** from the media objects **108** based on the zoomable ROI **112a**. The consecutive image **114** is displayed such that the media presentation **104** is created with the zoom-in effect applied on the zoomable ROI **112a**, upon presentation with image continuity sequence and semantic relevance between the first image **106** and the consecutive image **114**. The identification module **210**, the determining module **212**, and the retrieving module **214** is in communication with the generating module **216**.

[0049] In an embodiment, the generating module **216** may be configured to generate the consecutive image **114** based on the zoomable ROI **112a**. In an example, the generating module **216** may generate the consecutive image **114** upon failure of the retrieving module **214** to retrieve the consecutive image **114** from the media objects **108**. Further, the generating module **216** may be configured to embed the consecutive image **114** with the first image **106** for generating the media presentation **104**. In an embodiment, a graphics processor such as the graphics processing unit

(GPU) may include the generating module **216** and such graphics processor may be configured to perform working of the system **100** with respect to the generating module **216**.

[0050] FIG. **3** is a diagram illustrating an example process of identifying the ROIs **112**, e.g., ROI-1, ROI-2, ROI-3, ROI-n in the first image **106** by the identification module **210** of the system **100**, according to various embodiments.

[0051] In an embodiment, the identification module **210** of the system **100** may include Fast Region-based Convolutional Neural Network **302** (Fast-RCNN) for performing object detection techniques on the first image **106**. In an example, the identification module **210** may, for example, and without limitation, use a CNN based architecture such as VGG or ResNet followed by a pooling layer. The output of the pooling layer may be input to a fully connected layer to predict bounding boxes **304** around the identified ROIs **112** e.g., ROI-1, ROI-2, ROI-3, ROI-n in the first image **106**. Similarly, the ROIs **112** identification may be performed in the consecutive image **114** as well.

[0052] Once the ROIs **112** are identified, the identification module **210** may be configured to slice each of the ROI-1, ROI-2, ROI-3, ROI-n from the first image **106** and store in the database **102a**. A feature vector corresponding to each of the identified ROIs **112** is derived from an output layer of the Fast-RCNN **302**. In an example, the feature vector may be represented as: Let $R \in \{r.sub.1, r.sub.2 \dots r.sub.n\}$ for n number of ROIs **112**.

[0053] FIG. **4** is a diagram illustrating an example process of determining the zoomable ROI **112a** in the first image **106** by the determining module **212** of the system **100**, according to various embodiments.

[0054] In an embodiment, the determining module **212** may be configured to determine the zoomable ROI **112a** from the identified ROIs **112**. The determining module **212** may include a geometric feature score sub-module **402**, an image depth score sub-module **404**, a zoomable score sub-module **406**. In an example, the identified ROIs **112** are received as input by the determining module **212** for calculating such scores by the sub-modules and determining the zoomable ROI **112a** based on score.

[0055] In an example, the geometric feature score sub-module **402** may be configured to determine a geometric feature score based on shape of each of the ROIs **112** such as circle, ellipses, polygon, rectangle, triangle. The bounding box around each of the ROIs **112** may include object class and region information.

[0056] In an example, the image depth score sub-module **404** may be configured to determine an image depth in geometric selected object.

[0057] In an example, the zoomable score sub-module **406** may be configured to determine a zoomable score for each of the ROIs **112** based on the geometric feature score, the image depth score, and finally determine the zoomable ROI **112a**.

[0058] FIG. **5** is a diagram illustrating an example process for determining the geometric feature score for the ROIs **112** in the first image **106** by the geometric feature sub-module **402** of the system **100**, according to various embodiments.

[0059] In an embodiment, the geometric feature sub-module **402** is configured to determine a dimension for the ROIs **112**. A grid map **502** is determined for the ROIs **112** identified in the first image **106**. The grid map **502** is determined based on the dimension. In an example, the grid map **502** indicates partitioning each of the ROIs **112** into multiple grid segments such that the grid map **502** partition each of the ROIs **112** into a multiple set of pixels or a number of objects or a number of small images to fit into the grid map **502**. Further, the grid map **502** is mapped into multiple zones. The multiple zones may be indicative of a columnar location for each of the grid segments. For example, in one of the ROIs **112** the object being the window may be present at a certain grid segment of the grid map. Now to identify the grid segment for the window, the zone may be determined such that it indicates a coordinate or columnar location of the window.

[0060] In an embodiment, the geometric feature sub-module **402** is configured to determine a pixel

map **504** for each of the ROIs **112** based on the grid map **502**. In an example, the pixel map **504** may form a table including a pixel information of the zone corresponding to each of the ROIs **112**. In an example, the ROIs **112** is transformed into the pixel map of 24 bits retaining true colours. In the example, the pixel map **504** may correspond to the zone in which the window being the object in one of the ROIs **112** may be present.

[0061] The geometric feature sub-module **402** is configured to determine an association between the grid segments based on the grid map **502** and the pixel map **504**. The association may be indicative of distribution of geometric relationships within zones that relates to the grid map **502**. In an example, if the window being the object in one of the ROIs **112**, may be present in multiple zones of the grid segment then, the association between the grid segments is determined such that all multiple zones sharing common boundary edges are included in determining the pixel map **504** for the window.

[0062] The geometric feature sub-module **402** is configured to determine a pivotal coordinate **506** for each of the zones. In an example, the pivotal coordinate indicates a starting and an ending point of each of the zone in the grid map **502**. Such coordinates may also be considered for deducing a geometric features for identified ROIs **112**. The geometric feature score **506** is thus determined based on the grid map **502**, the pixel map **504**, the association between the plurality of grid segments, and the pivotal coordinate **506**. In an example, the geometric feature score **506** may associate relevance of the ROIs **112** with a shape of the object. In an example, the geometric feature score **506** of the window being the object in one of the ROIs **112** is relevant to shape such as circle, ellipses, polygon, rectangle, triangle, etc.

[0063] FIG. **6** is a diagram illustrating an example process for determining an image depth score **608** for the ROIs **112** by an image depth score sub-module **404** of the system **100**, according to various embodiments.

[0064] In an embodiment, the image depth score sub-module **404** may be configured to sample **602** the ROIs **112** to determine a geometry of each of the ROIs in the first image **106**. In an example, sample **602** is indicative of dividing each of the ROIs **112** into n number of parts or tiles to distinguish the geometry of the object. In an example, the window in one of the ROIs **112** may be divided into $N \times N$ tiles to preserve position of the window in the ROI.

[0065] The image depth score sub-module **404** may include a pretrained deep learning model **604** such as ResNet-50. The pretrained deep learning model **604** is used to extract feature vector from each of the ROIs **112**.

[0066] The image depth score sub-module **404** may be configured to determine a correlation image **605** and an image depth map **606** for the correlation image **605** based on the extracted features. In an example the correlation image **605** may be structurally equivalent to the ROIs **112**. In the example, K-Nearest Neighbor framework may be applied to determine the correlation image **114**. In the example, the correlation image **605** may be retrieved from the database **102a**.

[0067] The image depth score sub-module **404** may be configured to determine a correlation value for the correlation image **605**. In an example, one or more correlation images **605** are determined which be structurally equivalent to the ROIs **112** then only one of the correlation image **605** is selected with the correlation value higher than a pre-defined correlation coefficient. The analysis of correlation value may remove outliers from the determined one or more correlation images **605** and may further reduce the number of determined correlation image **605**.

[0068] In an embodiment, the image depth score sub-module **404** may be configured to determine the image depth map **606** for the ROIs **112**. In an example, the image depth map **606** for the ROIs **112** is determined corresponding to the depth map of the correlation image **605** which is structurally similar to the ROI. Thus, the image depth score **608** is determined based on the determined image depth map **606**. The image depth score **608** may signify the accuracy of the ROIs **112** with the selected correlation image **605**.

[0069] FIG. **7** is a diagram illustrating an example process for determining a zoomable score **702**

for the ROIs **112** by a zoomable score sub-module **406** of the system **100**, according to various embodiments.

[0070] In an embodiment, the zoomable score sub-module **406** is configured to determine the zoomable ROI **112a**, so as to remove other ROIs **112**, based on the geometric feature score **508** and the image depth score **608**. In an example, the zoomable score sub-module **406** may be a fusion decision model configured to receive the geometric feature score **508** (see, e.g., FIG. 5) and the image depth score **608**. The combination of the geometric feature score **508** and the image depth score **608** may be passed through a fully connected layer in the zoomable score sub-module **406** to determine the zoomable score **702** also classify the probable object class of the zoomable ROI **112a** thus selected based on the highest zoomable score **702**.

[0071] FIG. 8 is a diagram illustrating an example process for retrieving the consecutive image **114** by the retrieving module **214** of the system **100**, according to various embodiments.

[0072] In an embodiment, the retrieving module **214** is configured to search and retrieve consecutive image **114** from the database **102a** such as phone gallery in minimal time with high accuracy. In an example, the retrieving module **214** is configured to index **802** the media object **108** in the database **102a**. For example, Self-Organizing Map (SOM) may be used as a technique for indexing the media object **108**.

[0073] The retrieving module **214** is configured to prepare a cluster **804** for the media object **108**. The cluster **804** prepared with the media object **108** may signify that the media objects **108** present in the cluster **804** may have visually similar features with the first image **106**. The cluster **804** is indicative of classifying the media objects **108** into a given number of categories. The cluster **804** may use unsupervised machine learning techniques wherein the media object **108** with similar features are grouped together. In an example, k-means clustering, and spectral clustering may be used in the retrieving module **214** to prepare the cluster **804**. Thus, the retrieving module **214** is configured to determine an association between the zoomable ROI **112a** and the media object **108** based on the clustering **804**. In an example, the K-Nearest Neighbor supervised learning may be used to determine association between the at least one zoomable ROI and the plurality of media objects in each of the plurality of clusters.

[0074] The retrieving module **214** is configured to determine a similarity score **806** for each of the media objects **108** in cluster **804** based on the association and retrieve one of the media object **108** as the consecutive image **114** from the plurality of clusters based on the highest similarity score **806**.

[0075] Thus, the zoom-in effect is applied on the determined zoomable ROI **112a** and the consecutive image **114** determined from the database **102a** is displayed.

[0076] FIG. 9A is a diagram illustrating an example process for a dynamic global-local attention sub-module for determining a dynamic global local vector in the generating module **216** of the system **100**, according to various embodiments.

[0077] In an embodiment, the generating module **216** is configured to generate the consecutive image **114** based on the zoomable ROI **112a**. The consecutive image **114** is generated when the retrieving module **214** fails to retrieve the consecutive image **114** from the media objects **108** in the database **102a**. Further, the generating module **216** embed the generated consecutive image **114** with the first image **106** for generating the media presentation **104** such that the zoom-in effect is created on the zoomable ROI **112a** and subsequently the consecutive image **114** is displayed on the UE **102**.

[0078] In various embodiments, the generating module **216** includes the dynamic global-local attention sub-module **901**. The dynamic global-local attention sub-module **901** is configured to guide the generating module **216** in determining local and global saliency features of the zoomable ROI **112a** and the first image **106**. The dynamic global-local attention sub-module **901** may adjust the importance of local features such as features derived from the zoomable ROI **112a** and using the global features such as any semantic features derived from the first image **106**. In an example,

the dynamic global-local attention sub-module **901** may include a dual global-local attention processes to extract a ROI feature **902** and a semantic feature **904** respectively.

[0079] In an example, one of the dual global-local attention processes may be configured to extract from the ROI feature **902** to measure the importance of the first image **106**. In the example, the ROI feature **902** may be indicative of the object in the zoomable ROI **112a** such as the window. Similarly, another of the dual global-local attention processes may be configured to extract semantic feature **904** from the first image **106** to measure the importance of the ROI feature **902**. In the example, the semantic features **904** may be indicative of shape, color, size, texture of the object in the zoomable ROI **112a**.

[0080] In various embodiments, the dynamic global-local attention sub-module **901** is configured to concatenate the ROI feature **902** and the semantic feature **904** to determine a representation of the zoomable ROI **112a**. The representation thus determines a dynamic global-local attention vector **906**.

[0081] FIG. **9B** is a diagram illustrating an example process for generating the consecutive image **114** by the generating module **216** of the system **100**, according to various embodiments.

[0082] In an embodiment, the generating module **216** includes the dynamic global-local attention sub-module **901** and the context feature sub-module **907**. In an example, context may refer, for example, to the objects and concepts that appear in the image. Context may provide information about particular scene in image such as objects in image, their arrangement, relative physical size to other objects, and location. In the example, the context feature sub-module **907** may be a type of recurrent neural network configured to determine a context feature **908** from the first image **106**. In another example, the context feature sub-module **907** may be configured to determine the context feature **908** from a text provided as input by the user. In another instance, user's environmental context or environmental context in the first image may be determined as the context feature **908** such as a sunset, a rainfall. Thus, the context feature **908** may assist in generating multiple possibilities for the consecutive image **114**.

[0083] The dynamic global-local vector **906** and the context feature **908** are provided as input to a generator **909**. Further, the generating module **216** includes a generative adversarial network (GAN) architecture including the generator **909**, a semantic arbiter **910** and a discriminator **912** forming convolutional neural network for generation of the consecutive image **114**.

[0084] In an embodiment, the generator **909** may be a temporal generator neural network that models a transform function. In an example, the generator **909** may be configured to generate the consecutive image **114** based on the context feature **908** and the dynamic global-local attention vector **906**. Thus, the generated consecutive image **114** is contextually correct succeeding image in sequence with input image which in the instance is the first image **106**. It will be understood that input image varies at different instances such as the generated consecutive image **114** may become input image further in the process for which contextually correct succeeding image is to be generated. Thus, the generator **909** is configured to generate contextual image which is in sequence with input image.

[0085] In an embodiment, the semantic arbiter **910** may be configured to assess the generated consecutive image **114** in terms of quality and sequential semantics with respect to input image which in the present instance is the first image **106**. The semantic arbiter **910** may be configured to determine a contextual loss **911** by comparing sequential semantics of the generated consecutive image **114** with the first image **106**. Further in accordance with the contextual loss **911** the weights for the generator **909** are adjusted.

[0086] In an embodiment, the discriminator **912** is a classifier that may distinguish real data from the data created by the generator **909**. The discriminator **912** may be trained using real data instances or fake data instances created by the generator **909**. In an instance, the discriminator **912** is configured to distinguish between consecutive images originating from training dataset with generated consecutive images **114** to assess a loss function such as an adversarial loss **913**. In the

instance, the generator **909** and the discriminator **912** may use minimax GAN loss referring to the minimax simultaneous optimization of the discriminator and generator models. Further in accordance with the adversarial loss **913** the weights for the generator **909** are adjusted. [0087] Using the contextual loss **911** and the adversarial loss **913**, a combined loss function is derived and the generator **909** may be trained based on the combined loss function. Thus, the generator **909** may adjust weights in accordance with the contextual loss **911** and the adversarial loss **913** to generate the consecutive image **114**. Further, the generated consecutive image **114** may thus be displayed on the UE **102** such that the media presentation **104** creates the zoom-in effect upon presentation with the image continuity sequence and the semantic relevance.

[0088] FIG. **10** is a diagram illustrating an example use case for generating the media presentation **104**, according to various embodiments.

[0089] In various embodiments, the first image **106** may display a world map. The zoomable ROI **112a** is determined in the first image **106** e.g., a city in the world map. The zoom-in effect is created on the zoomable ROI **112a** e.g., the city illustrated as a bounding box. The consecutive image **114a** is displayed. The consecutive image **114a** illustrates a landscape scene in the city. Thus, creating a semantically, contextually relevant consecutive image **114**. The consecutive image **114a** may act as input image for the generating module **216** and the zoomable ROI **112a** e.g., a peak of mountain is determined in the consecutive image **114a**, which is illustrated with the bounding box. The zoom-in effect is created on the zoomable ROI **112a** in the consecutive image **114a** and the consecutive image **114b** is displayed. The consecutive image **114b** illustrates animals and plants which may be semantically, contextually relevant to previous image e.g., the consecutive image **114a**.

[0090] FIG. **11** is a flowchart illustrating an example method **1100** for generating the media presentation **104**, according to various embodiments. The method **1100** may be a computer-implemented method executed, for example, by the UE **102** and the modules **206**. For the sake of brevity, constructional and operational features of the system **100** that are already explained in the description of FIGS. **1**, **2A**, **2B**, **3**, **4**, **5**, **6**, **7**, **8**, **9A**, **9B** and **10** may not be repeated in the description of FIG. **11**.

[0091] At step **1102**, the method **1100** may include receiving the first image **106**, from the media object **108** stored in the database **102a**. The database **102a** may also include the pre-defined object **110** in the media object **108**.

[0092] At step **1104**, the method **1100** may include identifying the ROIs **112** in the first image **106** based on the pre-defined objects **110**.

[0093] At step **1106**, the method **1100** may include determining the zoomable ROI **112a** from the identified ROIs **112**. The zoomable ROI **112a** indicates one of the ROIs **112** with a zoom-in effect. The method **1100** may include determining the geometric feature score **508** for each of the ROIs **112**. The geometric feature score **508** is indicative of a geometric classification of each of the ROIs **112**. In an example, the method **1100** for determining the geometric feature score **508** may include determining a dimension of each of the ROIs **112**. The method **1100** may include determining the grid map **502** for each of the ROIs **112** based on the dimension. The grid map **502** may be indicative of partitioning the ROIs **112** into a plurality of grid segments. The method **1100** may include mapping the grid map **502** into the zones. The zones may be indicative of columnar location of each of the grid segments. The method **1100** may include determining the pixel map **504** for each of the ROIs **112** based on the grid map. In the method **1100** the pixel map **504** includes the pixel information of the zones corresponding to each of the ROIs **112**. The method **1100** may include determining the association between the grid segments based on the grid map **502** and the pixel map **504**. Further, the method **1100** may include determining a pivotal coordinate **506** for each of the zones, wherein the pivotal coordinate **506** indicates the starting and the ending point of each of the zones and the geometric feature score **508** is determined based on the grid map **502**, the pixel map **504**, the association between the plurality of grid segments, and the pivotal

coordinate **506**.

[0094] In an embodiment, the method **1100** may include determining the image depth score **608** for each of the ROIs **112**. The image depth score **608** is indicative of the image depth classification of each of the ROIs **112**. In an example, the first image **106** is sampled **602** to determine a position of the pre-defined objects in the first image **106**. The method **1100** may include extracting features from the sampled first image **106** using the pretrained machine learning model **604**. Further, the correlation images **605** are determined and the image depth map **605** is determined for each of the correlation images **605** based on the extracted features. In the example, the correlation image **605** is structurally equivalent to the first image **106**. The method **1100** may include determining a correlation value of each of the correlation images **605** to select at least one correlation image **605** with the correlation value higher than the pre-defined correlation coefficient. The method **1100** may include determining the image depth map **606** for the first image **106** based on the image depth map of the selected at least one correlation image **605** and further determining the image depth score **608** based on the determined image depth map **606**.

[0095] In an embodiment, the method **1100** may include determining the zoomable score **702** for each of the ROIs **112** based on the geometric feature score **508** and the image depth score **608**. The method **1100** may include determining the zoomable ROI **112a** based on the zoomable score **702**.

[0096] At step **1108**, the method **1100** may include retrieving the consecutive image **114** from the media objects **108** based on the zoomable ROI **112a**.

[0097] In an embodiment, for retrieving the consecutive image **114** from the database **102a** the method **1100** may include indexing **802** the media objects **108** stored in the database **102a**. Further, preparing the clusters **804** comprising of the media objects **108** such that the media objects **108** may be having visually similar features with that of the ROIs **112**. The method **1100** may include determining the association between the first image **106** and the media objects **108** in each of the clusters **804**. A similarity score is determined for each of the media objects **108** in the cluster **804** based on the association and one of the media object **108** is retrieved as the consecutive image **114** from the cluster **804** based on the similarity score **806**.

[0098] At step **1110**, the method **1100** may include generating the consecutive image **114** based on the zoomable ROI **112a** upon failure to retrieve the consecutive image **114** from the media object **108**.

[0099] In an embodiment, the method **1100** may include extracting the ROI feature **902** from the zoomable ROI **122a**. In an example, the ROI feature **902** is indicative of the object in the zoomable ROI **122a**.

[0100] The method **1100** may include extracting the semantic feature **904** from the first image **106**. In the example, the semantic feature **904** is indicative of shape, color, size, texture of the object in the zoomable ROI **112a**. The method **1100** may include concatenating the ROI feature **902** and the semantic feature **904** to determine the representation of the zoomable ROI **112a**. Further, the context feature **908** based on the first image **106** and the user's environmental context is determined. The dynamic global-local attention vector **906** is determined from the first image **106** based on the ROI feature **902** and the semantic feature **904**. The method **1100** may include generating the consecutive image **114** based on the context feature **908** and the dynamic global-local attention vector **906**.

[0101] In an embodiment, the method **1100** may include determining the generation loss through the semantic arbiter **910** by comparing the generated consecutive image **114** sequential semantics with the first image **106**.

[0102] At step **1112**, the method **1100** may include embedding the consecutive image **114** with the first image **106** for generating the media presentation **106**. In an embodiment, the zoom-in effect is created on the zoomable ROI **112a**. Further, the consecutive image **114** is displayed on the UE **102** such that the media presentation **104** creates the zoom-in effect upon presentation with the image continuity sequence and the semantic relevance.

[0103] The disclosure provides various example, non-limiting, advantages: [0104] The disclosure enables the user to create engaging content for viewing. [0105] The disclosure provides zooming transition ability in a slideshow of images. [0106] The disclosure includes using machine learning techniques to retrieve images from the database which may be semantically and contextually relevant with respect to input image. Thus, manual efforts and time of the content creator is saved. [0107] The disclosure includes using machine learning techniques to generate semantically and contextually relevant with respect to input image. Thus, the content creator may not have to be dependent on manual sorting, searching.

[0108] While specific language has been used to describe the disclosure, any limitations arising on account of the same are not intended. As would be apparent to one skilled in the art, various working modifications may be made to the method in order to implement the concept as taught herein.

[0109] The drawings and the forgoing description give examples of various embodiments. Those skilled in the art will appreciate that one or more of the described elements may well be combined into a single functional element. Various elements may be split into multiple functional elements. Elements from one embodiment may be added to another embodiment. For example, orders of processes described herein may be changed and are not limited to the manner described herein.

[0110] While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. A method for generating a media presentation, the method comprising: receiving a first image, from a plurality of media objects comprising a plurality of specified objects; identifying a plurality of regions of interest (ROIs) in the first image based on the specified objects; determining at least one zoomable ROI from the identified plurality of ROIs, wherein the at least one zoomable ROI includes one of the plurality of ROIs with a zoom-in effect; retrieving a consecutive image from the plurality of media objects based on the at least one zoomable ROI; generating the consecutive image based on the at least one zoomable ROI based on a failure to retrieve the consecutive image from the plurality of media objects; and embedding the consecutive image with the first image to generate the media presentation.
2. The method as claimed in claim 1, comprising generating the media presentation using a convolutional neural network (CNN).
3. The method as claimed in claim 1, wherein determining the at least one zoomable ROI comprises: determining a geometric feature score for each of the plurality of ROIs, wherein the geometric feature score is indicative of a geometric classification of each of the plurality of ROIs; determining an image depth score for each of the plurality of ROIs, wherein the image depth score is indicative of an image depth classification of each of the plurality of ROIs; determining a zoomable score for each of the plurality of ROIs based on the geometric feature score and the image depth score; and determining the at least one zoomable ROI based on the zoomable score.
4. The method as claimed in claim 3, wherein determining the geometric feature score comprises: determining a dimension of each of the plurality of ROIs; determining a grid map for each of the plurality of ROIs based on the dimension, wherein the grid map indicates partitioning each of the plurality of ROIs into a plurality of grid segments; mapping the grid map into a plurality of zones, wherein the plurality of zones is indicative of columnar location of each of the plurality of grid segments; determining a pixel map for each of the plurality of ROIs based on the grid map, wherein

the pixel map comprising pixel information of the plurality of zones corresponding to each of the plurality of ROIs; determining an association between the plurality of grid segments based on the grid map and the pixel map, determining a pivotal coordinate for each of the plurality of zones, wherein the pivotal coordinate indicates a starting and an ending point of each of the plurality of zones; and determining the geometric feature score based on the grid map, the pixel map, the association between the plurality of grid segments, and the pivotal coordinate.

5. The method as claimed in claim 3, wherein determining the image depth score comprises: sampling the first image to determine a position of each of the plurality of specified objects in the first image; extracting features from the sampled first image using a trained machine learning model; determining a plurality of correlation images and an image depth map of each of the plurality of correlation images based on the extracted features, wherein the plurality of correlation images is structurally equivalent to the first image; determining a correlation value of each of the plurality of correlation images to select at least one correlation image with the correlation value higher than a specified correlation coefficient; determining the image depth map for the first image based on the image depth map of the selected at least one correlation image; and determining the image depth score based on the determined image depth map.

6. The method as claimed in claim 1, wherein retrieving the consecutive image from the plurality of media objects comprises: indexing the plurality of media objects stored in a database; preparing a plurality of clusters comprising of the plurality of media objects, based on the plurality of media objects with visually similar features; determining an association between the first image and the plurality of media objects in each of the plurality of clusters; determining a similarity score for each of the plurality of media objects in each of the plurality of clusters based on the association; and retrieving one of the plurality of media objects, as the consecutive image from the plurality of clusters based on the similarity score.

7. A system for generating a media presentation, the system comprising: at least one processor, comprising processing circuitry, individually and/or collectively, configured to: receive a first image, from a plurality of media objects comprising a plurality of specified objects; identify a plurality of regions of interest (ROIs) in the first image based on the specified objects; determine at least one zoomable ROI from the identified plurality of ROIs, wherein the at least one zoomable ROI includes one of the plurality of ROIs with a zoom-in effect; retrieve a consecutive image from the plurality of media objects based on the at least one zoomable ROI; generate the consecutive image based on the at least one zoomable ROI based on a failure to retrieve the consecutive image from the plurality of media objects; and embed the consecutive image with the first image to generate the media presentation.

8. The system as claimed in claim 7, wherein at least one processor, individually and/or collectively, is configured to generate the media presentation using a convolutional neural network (CNN).

9. The system as claimed in claim 7, wherein at least one processor, individually and/or collectively, is configured to: determine a geometric feature score for each of the plurality of ROIs, wherein the geometric feature score is indicative of a geometric classification of each of the plurality of ROIs; determine an image depth score for each of the plurality of ROIs, wherein the image depth score is indicative of an image depth classification of each of the plurality of ROIs; determine a zoomable score for each of the plurality of ROIs based on the geometric feature score and the image depth score; and determine the at least one zoomable ROI based on the zoomable score.

10. The system as claimed in claim 9, wherein at least one processor, individually and/or collectively, is configured to: determine a dimension of each of the plurality of ROIs; determine a grid map for each of the plurality of ROIs based on the dimension, wherein the grid map indicates partitioning each of the plurality of ROIs into a plurality of grid segments; map the grid map into a plurality of zones, wherein the plurality of zones is indicative of columnar location of each of the

plurality of grid segments; determine a pixel map for each of the plurality of ROIs based on the grid map, wherein the pixel map comprising of a pixel information of the plurality of zones corresponding to each of the plurality of ROIs; determine an association between the plurality of grid segments based on the grid map and the pixel map, determine a pivotal coordinate for each of the plurality of zones, wherein the pivotal coordinate indicates a starting and an ending point of each of the plurality of zones; and determine the geometric feature score based on the grid map, the pixel map, the association between the plurality of grid segments, and the pivotal coordinate.

11. The system as claimed in claim 9, wherein at least one processor, individually and/or collectively, is configured to: sample the plurality of ROIs to determine a geometry of each of the plurality of ROIs in the first image, wherein the sample is indicative of dividing each of the plurality of ROIs into n number of parts; extract features from each of the sampled plurality of ROIs using a trained machine learning model; determine a plurality of correlation images and an image depth map for each of the plurality of correlation images based on the extracted features, wherein the plurality of correlation images is structurally equivalent to the plurality of ROIs; determine a correlation value of each of the plurality of correlation images to select at least one correlation image with the correlation value higher than a specified correlation coefficient; determine the image depth map for each of the plurality of ROIs based on the image depth map of the selected at least one correlation image; and determine the image depth score based on the determined image depth map.

12. The system as claimed in claim 7, wherein at least one processor, individually and/or collectively, is configured to: index the plurality of media objects stored in a database; prepare a plurality of clusters comprising of the plurality of media objects, based on the plurality of media objects with visually similar features with the first image; determine an association between the at least one zoomable ROI and the plurality of media objects in each of the plurality of clusters; determine a similarity score for each of the plurality of media objects in each of the plurality of clusters based on the association; and retrieve one of the plurality of media objects as the consecutive image from the plurality of clusters based on the similarity score.

13. The system as claimed in claim 7, wherein at least one processor, individually and/or collectively, is configured to: extract a plurality of zoomable ROI features from the at least one zoomable ROI, wherein the plurality of zoomable ROI features is indicative of an object in the at least one zoomable ROI; extract a plurality of semantic features from the first image, wherein the plurality of semantic features is indicative of shape, color, size, texture of the object in the at least one zoomable ROI; concatenate the plurality of semantic features and the plurality of zoomable ROI features to determine a representation of the at least one zoomable ROI; determine a context feature from the first image based on the first image and an environmental context; determine a dynamic global-local attention vector from the first image based on the plurality of semantic features and the plurality of ROI features; generate the consecutive image based on the context feature and the dynamic global-local attention vector.

14. The system as claimed in claim 13, wherein at least one processor, individually and/or collectively, is configured to: determine generation loss via a semantic arbiter, by comparing a sequential semantics of the generated consecutive image with the first image.

15. The system as claimed in claim 7, wherein at least one processor, individually and/or collectively, is configured to: create a zoom-in effect on the at least one zoomable ROI; display the consecutive image such that the media presentation creates the zoom-in effect upon presentation with an image continuity sequence and a semantic relevance.
